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Latent Transformation-Equivariance for Spatial Reasoning in a Vision–Language Model: Algebra Compliance without Downstream Benefit

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Abstract

Can training an answer-relevant object-pair spatial state in a vision–language model (VLM) to obey a predeclared, geometry-derived transformation algebra deliver downstream benefit beyond matched augmentation, output-consistency, and invariance controls? We test this with a one-seed falsification pilot on synthetic plan-view scenes rendered for Qwen3-VL-8B-Instruct, with a verified implementation: the equivariance loss couples the pair state on the original and transformed images, its magnitude is logged and checked per epoch (a manipulation check that caught and voided an earlier zero-loss implementation), and six matched arms share one answer path and differ only in data/loss. The result is a clean separation. When the loss is actually applied, the latent complies with the predeclared algebra: on a held-out transform composition never seen in training, EquiOrient's equivariance error is 0.0452 vs. 14.6516 for augmentation-only (a a $\approx$324$\times$ reduction), and the per-transform correct-vs-wrong contrast discriminates sharply (0.0331 vs. 4.8945 on the horizontal reflection), with a wrong-geometry control that obeys its own (incorrect) law per-transform. Yet no downstream benefit appears: the behavioral composition test is uninformative by construction (the task reduces to a binary left/right axis on which the answer algebra is closed, so every arm reaches ceiling), and a held-out relation-family probe (depth) is flat across arms (0.53–0.55). The paper is a mechanistic negative: representation-level algebra compliance is achievable and measurable, but it did not transfer to behavior or to an unseen relation family within Phase-1 scope.

1. Introduction

Answer-level consistency laws are a standard diagnostic: a model that flips its answer when an image is mirrored is "using" the visual evidence [8]. But a model can obey an answer law while its representation transforms arbitrarily, and vice versa. EquiOrient tests the stronger object: can a typed, answer-relevant spatial latent be trained to obey a predeclared, geometry-derived transformation algebra, and does that compliance pay off behaviorally or in transfer to unseen structure?

This paper's central contribution is the honest answer to both parts, obtained with a discipline that the first attempt lacked. Our first implementation—and, unknowingly, every GPU run built on it—had identically zero structural losses: the equivariance term fabricated $\rho(T)\mathbf{z}$ from the same $\mathbf{z}$ instead of coupling $\mathbf{z}(x)$ and $\mathbf{z}(Tx)$ from the original and transformed images. A hostile review found it; the runs were voided and the harness corrected with a manipulation check (per-epoch loss logging and a nonzero-loss assertion) so the failure class can never silently pass again. The corrected pilot is the first valid test, and it produced a sharply interpretable result: the equivariance objective works at the representation level and fails to deliver downstream benefit in the tested scope.

Concretely, on a held-out composition $V \circ H$ never seen in training, EquiOrient's latent equivariance error is 0.0452 against 14.6516 for augmentation-only and 0.0261 for the wrong-geometry control—yet the wrong-geometry control obeys its own incorrect law per-transform (0.0198 on H), while EquiOrient violates that wrong law (4.8945). The correct-versus-wrong contrast is therefore measured, not assumed. Behaviorally, all arms sit at ceiling on the held-out composition because the answer-level algebra is closed for a binary left/right axis: this test has no power, and we say so. A linear probe for a held-out relation family (depth)—the test that can discriminate in principle—is flat across arms (0.53–0.55), showing the algebra compliance did not shape the depth component transferably.

We release the protocol, harness, committed matri-

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ces, and a serverless Modal pipeline that reproduces the full six-arm pilot in minutes.

Our contributions are:

- a manipulation-verified six-arm falsification harness for transformation-equivariance training, including the per-epoch structural- loss check that caught and voided an identically-zero implementation;
- the first valid measurement of the question: EquiOrient achieves strong, specific, two-regime latent algebra compliance (including on a held-out composition) without any downstream benefit in scope;
- per-transform and wrong-law causal metrics that separate correct from wrong actions where composition metrics cannot;
- released artifacts: protocol, harness, matrices, and a pay-per- second Modal pipeline (full pilot $\approx$20 min, $<$2).

2. Related work

The novelty gate (frozen 2026-08-13) screened 30 close or foundational works; the top threats and our differentiation:

Latent group representations. The Homomorphism AutoEncoder (HAE) [7] learns group-structured latent representations with homomorphism-based composition from observed transitions. EquiOrient does not claim to learn an action group—the geometry action is predeclared—and its question is VLM-specific: whether a constrained spatial action on an answer-relevant pair state improves compositional spatial reasoning beyond matched behavioral baselines.

Representation-level geometric supervision. GASP (Beyond 3D VQAs) [5] injects 3D spatial priors into VLMs via correspondence invariance and depth consistency. EquiOrient instead asks whether relation-relevant components should transform non-uniformly under the same intervention, and whether the composed algebra transfers to unseen transforms—with a wrong-action causal control GASP does not have.

Transformation-pair consistency. SAGE [9] trains VLMs to stay logically consistent under paired geometric and linguistic operations (output/reward-level). EquiOrient treats such consistency as a matched baseline and asks whether enforcing the structure on the answer-path state yields additional generalization; our output-consistency arm is exactly SAGE-style training.

Consistency vs. accuracy. Consistent Yet Wrong [1] shows multi-view consistency can coexist with wrong answers (static analysis of pretrained models). We perform a controlled fine-tuning transition and measure the equivariance axis as a training outcome; our result—algebra compliance without transfer—extends their static finding to a causal setting.

Equivariant representation learning (Cohen and Welling [3], Bronstein et al. [2]) is foundational; EquiOrient's contribution is the VLM answer-path instantiation with a typed heterogeneous algebra and matched controls. Additional screened neighbors (chain-of-spatial-thought, 3DThinker, GeoWorld-VLM) [4, 6, 10] lack the predeclared pair-state algebra or its compositional test.

3. Method

3.1. Typed spatial state and algebra

The pair encoder maps pooled deepstack features of two objects to a 512-dim typed latent $\mathbf{z} = [\mathbf{z}_h; \mathbf{z}_v; \mathbf{z}_d; \mathbf{z}_{\text{orient}}]$ (4×128); the relation head is $W_{\text{rel}}[\mathbf{z}_h; \mathbf{z}_v]$ with forced decoding to left/right/above/below. Predeclared actions (geometry only, never the relation label): $\rho(H)$ flips $\mathbf{z}_h$, $\rho(V)$ flips $\mathbf{z}_v$, $\rho(V \circ H) = \rho(V)\rho(H)$, depth and orientation invariant. The algebra is verified executably (21 unit tests; 10,880 law checks on the scene pack, 0 violations; hostile renderer-bug test).

3.2. Six matched arms

All arms share the identical answer path and trainable set (vision LoRA on fused qkv/proj/c_fc/c_proj; text backbone and lm_head frozen) and start from a common initialized state restored per arm. Arms: (1) ordinary SFT/LoRA, original images only; (2) augmentation-only, H/V pairs, answer loss; (3) output-consistency, answer loss + soft answer-law consistency $\text{MSE}(\text{logits}_{Tx}, \text{perm}_{\text{law}(T)}(\text{logits}_x))$ with detached law targets; (4) latent-invariance, answer loss + $\|\mathbf{z}(x) - \mathbf{z}(Tx)\|^2$; (5) EquiOrient, answer loss + $\|\rho(T)\mathbf{z}(x) - \mathbf{z}(Tx)\|^2$; (6) wrong-geometry, the same form with the wrong action (H acting on $\mathbf{z}_v$). Structural losses couple the identity-image state $\mathbf{z}(x)$ with the transform-image state $\mathbf{z}(Tx)$ on the same object ids (identity features computed once per scene per step, gradients preserved). The structural loss magnitude is logged per epoch and asserted nonzero (manipulation check). The $\lambda \in \{0.1, 1.0, 10.0\}$ grid is shared by all structural arms and selected on the fixed validation slice only; wrong-geometry uses EquiOrient's selected λ.

3.3. Held-out tests (single, post-selection, per arm)

(i) behavioral composition $V \circ H$ on hold-out scenes; (ii) the same with $\mathbf{z}$ zeroed (causal ablation; the head's bias decode on balanced binary labels is 0.5 by construction—we report it as a plumbing check, not chance); (iii) paired both-correct; (iv) latent equivariance error $\|\rho(V \circ H)\mathbf{z}(x) - \mathbf{z}(Tx)\|^2$ plus per-transform errors under the correct and the wrong rho (the compo-

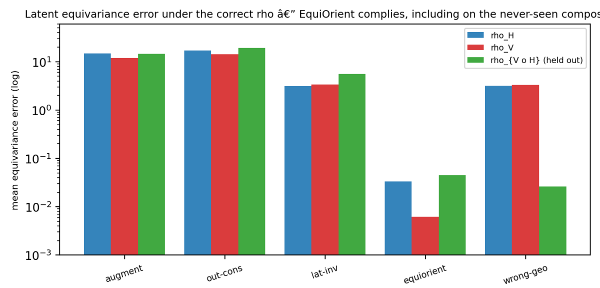

Figure 1. Latent equivariance error under the correct rho, per transform (log scale). EquiOrient complies on the seen reflections and on the never-seen composition; the wrong-geometry control's small composition bar reflects its wrong law composing to the correct action on $V \circ H$ (Fig. 2); the remaining controls minimize nothing.

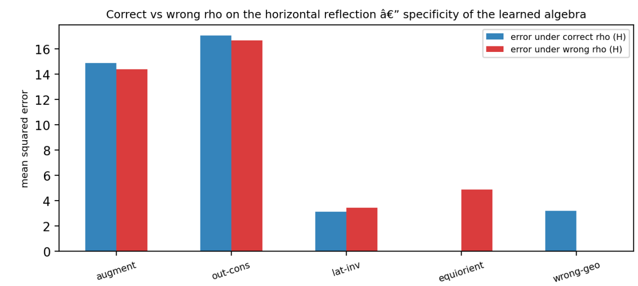

Figure 2. Specificity: errors under the correct vs. wrong rho on the horizontal reflection. EquiOrient minimizes the correct law and violates the wrong one; the wrong-geometry control is the mirror image.

sition metric alone cannot separate correct from wrong rho: the axis-swapped wrong laws compose to the correct action on $V \circ H$); (v) the depth probe: a logistic regression on $\mathbf{z}$ (full and $\mathbf{z}_d$-only) fit on validation-scene depth pairs, scored on hold-out-scene depth pairs.

4. Protocol

Plan-view synthetic scenes (320×320), margin-guaranteed strict relations, distinct depths, 5 objects (3 rects + 2 lines); 17 scenes split 10/4/3 by scene id; identity, H, V seen in training; $V \circ H$ never. Qwen3-VL-8B-Instruct at a frozen revision, transformers==4.57.6, bf16/sdpa; AdamW lr 1e-4, batch 8, 2 epochs, gradient checkpointing; one-seed Phase-1 falsification scope. The task is effectively binary: every ordered pair carries a strict horizontal relation, so the label rule assigns left/right and never above/below—we disclose this and report the behavioral composition test as uninformative by construction.

5. Results

Table 1 reports the corrected pilot (Modal L40S, 1357 s). Manipulation check passed: structural losses are nonzero in every structural arm (the manipulation check); they decrease for EquiOrient and output-consistency, and rise for latent-invariance, whose invariance objective competes with the answer loss (EquiOrient: 0.015562 → 0.013168; output-consistency: 0.179039 → 0.161825), and the answer loss converges (EquiOrient: 7.4582 → 1.2425).

Representation-level algebra compliance (the positive mechanistic result). EquiOrient's latent obeys the predeclared algebra: per-transform equivariance error 0.0331 (H), 0.0062 (V), and 0.0452 on the never-seen composition—versus 14.9–19.2 for augmentation-only and output-consistency (latent-invariance: 3.1–5.6, still ∼70–450× above EquiOrient). The wrong-geometry control is the clean causal contrast: it obeys its own incorrect law per-transform (0.0198 on H, 0.0052 on V) and violates the correct law there (3.1830), while on the composition its composed wrong law coincides with the correct action (0.0261), exactly as the axis-symmetry analysis predicts (Fig. 2). The correct-vs-wrong contrast is therefore measured and specific.

No downstream benefit (the negative result). The behavioral composition test is at ceiling for every arm (1.0000)—uninformative by construction on the binary axis, as disclosed. Validation accuracy at the selected λ (Table 2) is 0.98–0.99 for every causal arm in both regimes, with grid accuracy nearly flat across λ (EquiOrient: 0.9792/0.9833/0.9958 at 0.1/1.0/10.0), so λ selection on the 4-scene validation slice is noisy; we report it and never tune on the holdout. The depth probe is flat: full-z 0.53–0.55 and $\mathbf{z}_d$-only 0.52–0.55 across all arms (EquiOrient: 0.5333 full / 0.5333 $\mathbf{z}_d$), all within binomial noise of chance at n=60 hold-out pairs (Fig. 3). Both-correct is 1.0000 everywhere; the causal ablation (zeroing $\mathbf{z}$) drops every arm to 0.5000, the bias decode—a plumbing check that passed.

Replication in the 4-object regime (Regime A). The same corrected harness on the original 4-object scenes reproduces the pattern: EquiOrient's per-transform errors are 0.1199 (H), 0.1054 (V), and 0.1717 on the held-out composition versus 10.3584 for augmentation-only (a ≈60× reduction); the wrong-geometry control again obeys its own law (0.1118 on H) while EquiOrient violates it, and the depth probe is again within binomial noise (n=36; EquiOrient full-z 0.5000 vs. augmentation 0.6111). The mechanistic result is stable across both scene regimes.

arm	ρ_H	ρ_V	$\rho_{V\circ H}$	wrong-ρ_H
augmentation-only	14.8577	11.9853	14.6516	14.3586
output-consistency	17.0195	14.1305	19.2160	16.6655
latent-invariance	3.1233	3.4106	5.5740	3.4288
EquiOrient	0.0331	0.0062	0.0452	4.8945
wrong-geometry	3.1830	3.3059	0.0261	0.0198

Table 1. Latent equivariance error (mean $\|\rho(T)\mathbf{z}(x) - \mathbf{z}(Tx)\|^2$) on hold-out scenes under the correct rho ($\rho_H, \rho_V, \rho_{V\circ H}$) and the wrong rho. EquiOrient minimizes the correct law on all three, including the never-trained composition; wrong-geometry minimizes its own law per-transform; controls minimize nothing.

arm	val	hold-out $V \circ H$	z-corr	both	probe
augmentation-only	0.9917	1.0000	0.5000	1.0000	0.5500
output-consistency	0.9917	1.0000	0.5000	1.0000	0.5500
latent-invariance	0.9792	1.0000	0.5000	1.0000	0.5500
EquiOrient	0.9958	1.0000	0.5000	1.0000	0.5333
wrong-geometry	0.9917	1.0000	0.5000	1.0000	0.5333

Table 2. Behavioral metrics at the selected λ (Regime B). z-corr = accuracy with $\mathbf{z}$ zeroed (bias decode, not chance). The behavioral columns are uninformative by construction on the binary task; the probe column is flat.

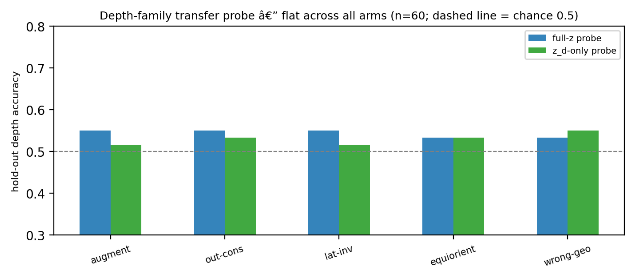

Figure 3. Held-out depth-family probe: flat across all arms, full-z and $\mathbf{z}_d$-only. Dashed line: chance (0.5).

Interpretation. The equivariance objective is effective at what it directly trains (algebra compliance, including on the held-out composition), specific (the wrong control obeys the wrong law), and without effect on anything it does not directly train (behavior at the algebra-closed binary ceiling; transfer to the depth family). This is the mechanistic negative the design was built to find: representation fidelity improved, downstream behavior did not follow.

6. Discussion

The corrected design turns the earlier null into an interpretable mechanistic finding. Three points deserve emphasis.

The composition test alone cannot validate an algebra. Because the wrong law is the axis-swapped law and $V \circ H$ is symmetric under the swap, both correct and wrong training compose to the same action on the composition (0.045 vs. 0.026). Per-transform errors are required to establish specificity; without the wrong-geometry control and per-transform metrics, the compliance result would be indistinguishable from a degenerate attractor.

Algebra closure bounds the behavioral test. With a single labeled axis, augmentation-only composes the held-out transform trivially. Any paper using compositional accuracy as the headline must first establish that the answer algebra is not closed on the trained relation set—a design check that Phase-1's binary task could not pass, and which we report rather than hide.

Scope of the negative. One seed, two epochs, synthetic plan-view scenes, one backbone. The depth probe's n=60 gives limited power (95% CI for 0.5333: 0.41–0.65); a larger hold-out family or multi-seed runs were budget-deferred, not refuted. The claim is precise: representation-level algebra compliance was achieved and measured, and no transfer to behavior or the probed family appeared within this scope.

7. Conclusion

With a manipulation-verified implementation, we find that training an answer-path latent in a VLM to obey a predeclared geometry-derived transformation algebra achieves strong, specific algebra compliance (including on a held-out composition, ≈324× below the augmentation-only and output-consistency controls), while delivering no measurable downstream benefit on the algebra-closed behavioral test or a held-out relation family probe. The full protocol, harness, committed matrices, and a serverless Modal pipeline are released; the design is re-testable in harder regimes in minutes and at cents.

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A. Full result matrices and ledger

Complete per-arm matrices (validation, held-out, z-corrupted, both-correct, latent error, per-transform rho errors under correct and wrong laws, depth probes, per-epoch loss curves, lambda selections) are committed at results/equiorient/..._final_matrix.json. Earlier GPU attempts were voided per protocol (implementation deviations, logged; the zero-structural-loss failure is documented in the decision log and excluded from all tables). Compute: 1×L40S (Modal, 1357 s; repo commit 65c6c1c9). Reproducibility: harness (scripts/equiorient/pilot_harness.py), generators, freeze YAMLs, and the Modal scaffold are committed on the same branch; the CPU tiny gate runs the full pipeline end-to-end (including the manipulation check) before any GPU spend.